

AWS IAM Investigation Lab Report

 **Name:**

Rituraj Pratap Singh

This lab walks through creating a limited-permission IAM user, generating login/S3 activity, inspecting permissions, and analyzing user actions with CloudTrail.

1. Overview

In this lab, you will:

- Create an IAM user
 - Investigate their permissions
 - Review what services they accessed
 - Correlate their actions with CloudTrail
-

2. Prerequisites

- AWS Console access with admin rights
 - CloudTrail **Event history** enabled (default for most accounts)
-

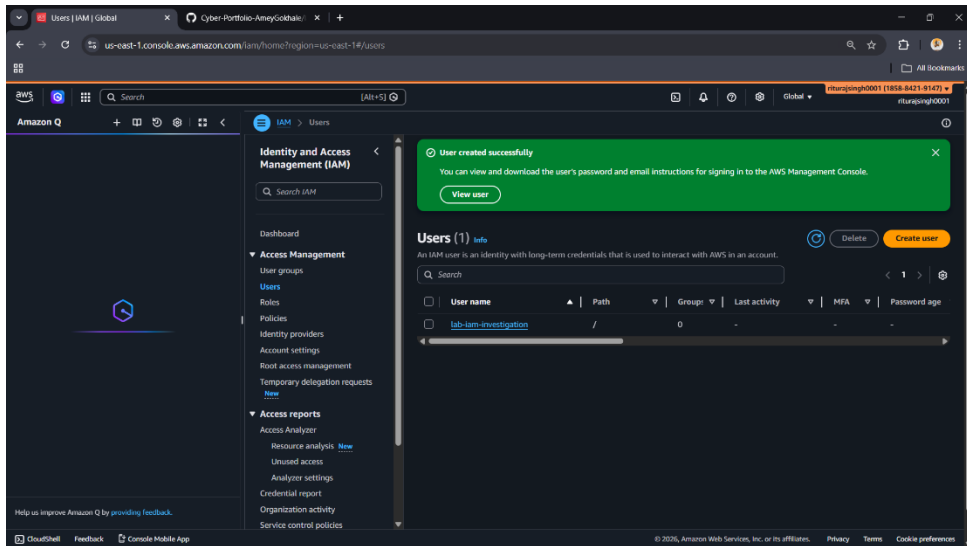
3. Steps

Step 1 – Create Investigation User

1. Open **IAM** in AWS Console.
2. In the left menu → **Users** → **Create user**.
3. Set user name: lab-iam-investigation → **Next**.
4. Permissions:
 - Choose **Attach policies directly**
 - Search and select: **AmazonS3ReadOnlyAccess**
5. Click **Next** → **Create user**.

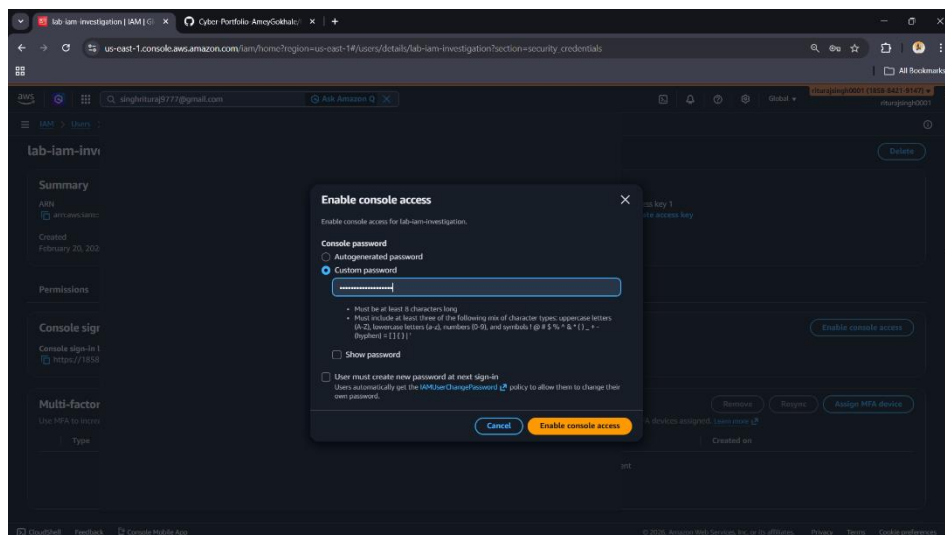
6. Open the user page.

Screenshot I1 – IAM User Created



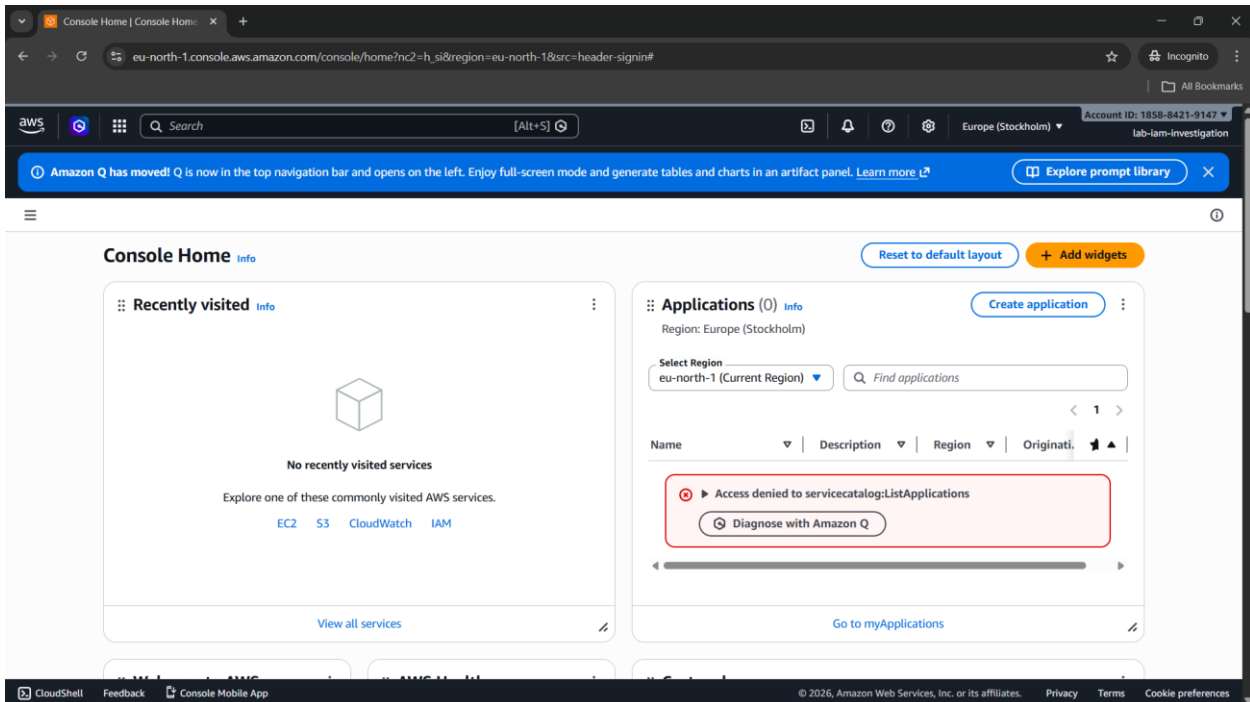
Step 2 – (Optional) Generate Console Login Activity

1. On the lab-iam-investigation user page → **Security credentials**.
2. Under Console password → **Manage console access**.
3. Enable **Console access** and set a custom password.



4. Open an **Incognito/Private Window**.

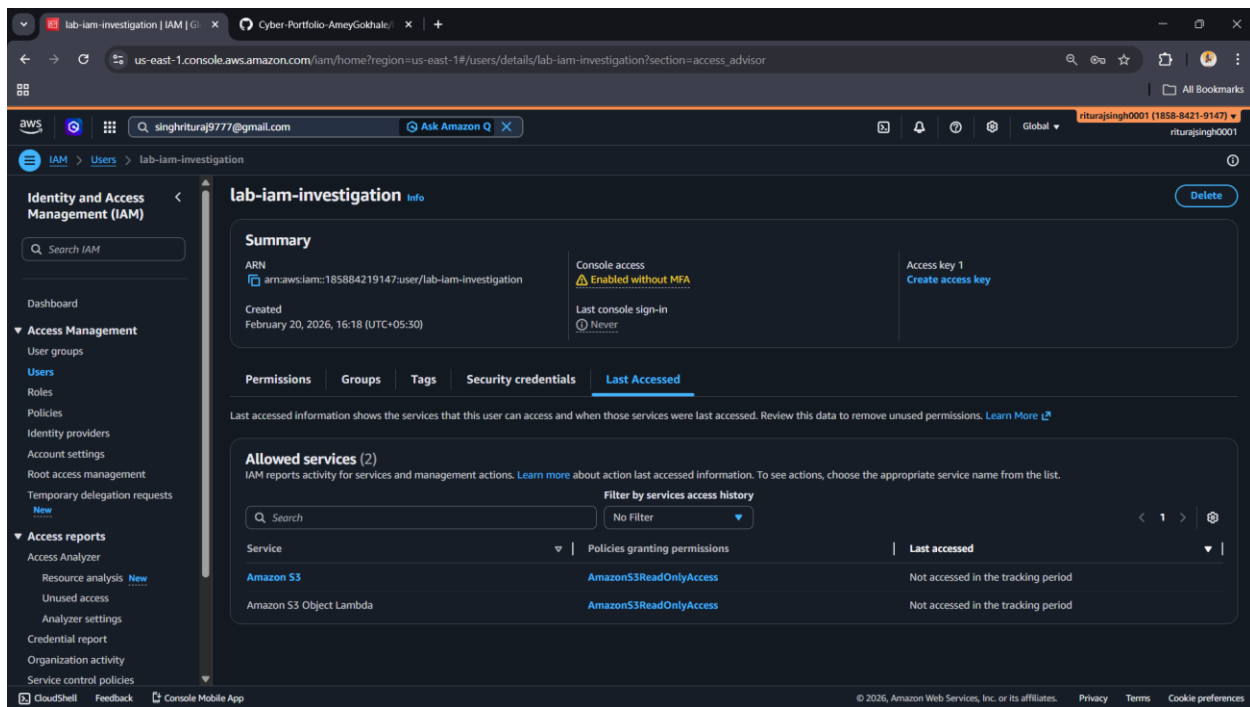
5. Log in using the IAM user credentials.
6. Navigate to **S3**, view bucket list, maybe open a bucket.



7. Log out.

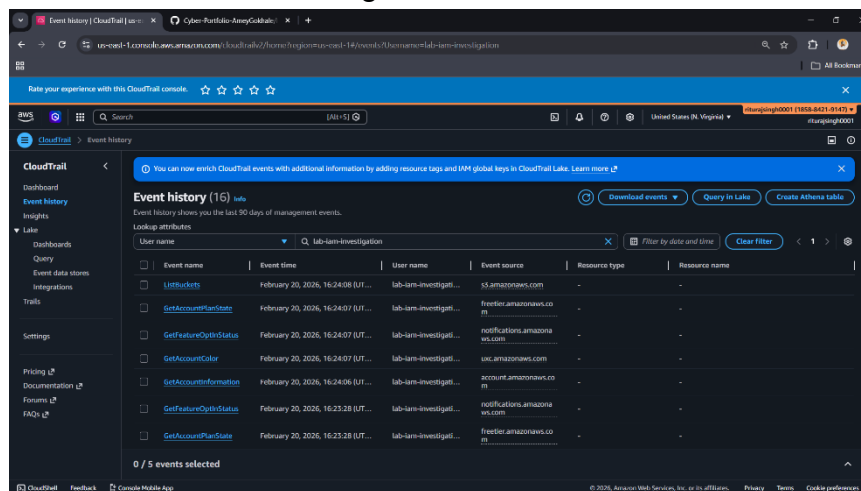
Step 3 – Inspect User in IAM (Permissions & Access Advisor)

1. Return to your admin session.
2. Open **IAM** → **Users** → **lab-iam-investigation**.
3. Go through:
 - **Permissions** tab → confirm attached policies
 - **Groups** tab → confirm no groups (unless added)
 - **Access Advisor** tab → shows last accessed services



Step 4 – Correlate Activity with CloudTrail

1. Open **CloudTrail** in AWS Console.
2. Left menu → **Event history**.
3. Set:
 - **Lookup attribute:** Username
 - **Value:** lab-iam-investigation

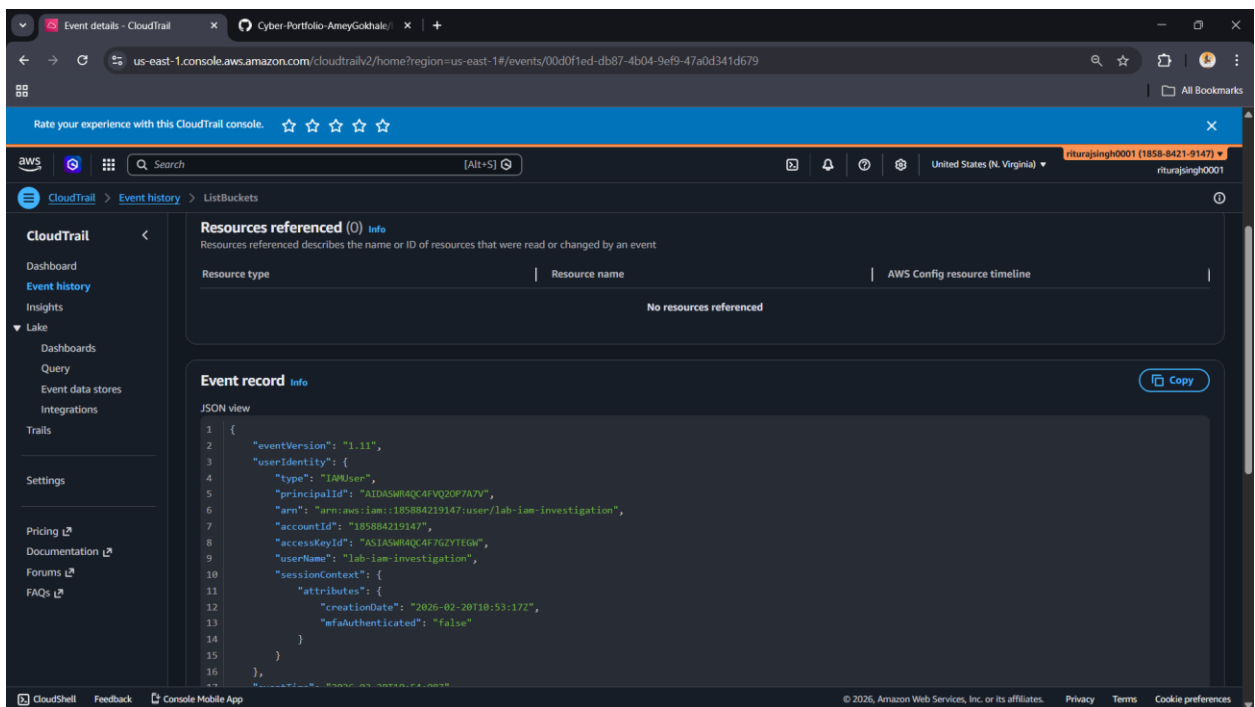


4. View events such as:

- ConsoleLogin
- ListBuckets
- GetObject

5. Click an event to view:

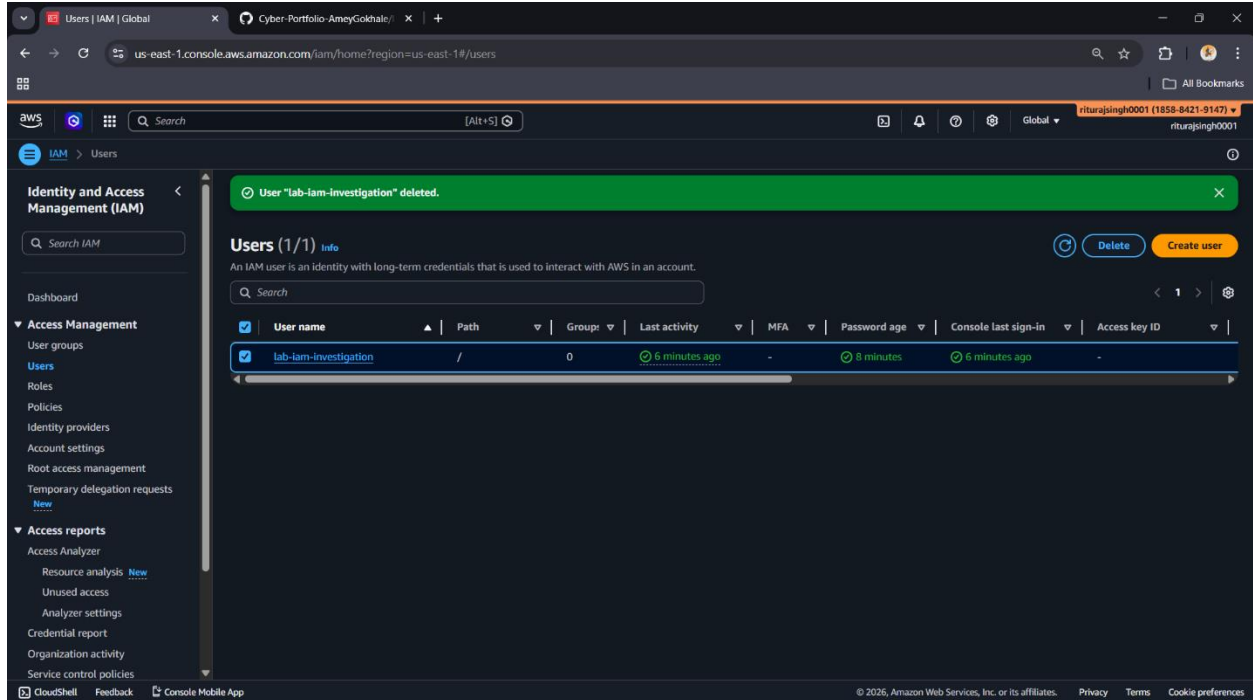
- eventTime
- sourceIPAddress
- userIdentity.arn
- requestParameters



4. Cleanup

1. Go to **IAM** → **Users**.
2. Select lab-iam-investigation.

3. Click **Delete user** → confirm.



5. Conclusion

This lab provided hands-on experience in cloud security and user activity investigation. It demonstrated how organizations can monitor and secure their cloud environments using AWS services like IAM and CloudTrail.

Understanding these concepts is crucial for roles like **Security Analyst**, **SOC Analyst**, and **Cloud Security Engineer**.